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A monsoon-season precipitation $\delta^{18}O$ isoscape for India: Insights into moisture recycling and atmospheric transport

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Highlights:
1. During the Indian summer monsoon, precipitation $\delta^{18}O$ values exhibit a progressive decrease in a southwest-to-northeast direction.
2. The spatial variability pattern aligns more closely with recycled rainfall than rainfall variability itself.
3. This observation offers potential for reinterpreting the amount effect from a new perspective.

Abstract

The spatial distribution of precipitation stable isotopes ("isoscapes") provides important insights into atmospheric moisture transport, land–atmosphere interactions, and regional hydrological processes. In this study, we construct a monsoon-season precipitation oxygen isotope ($\delta^{18}O$) isoscape for India and investigate the mechanisms governing its spatial variability. Despite the limitations imposed by the sparse observational network, the reconstructed isoscape reveals a pronounced southwest-to-northeast isotopic gradient, with relatively enriched values over southwestern India and progressively depleted values toward northeastern India. While an inverse relationship between rainfall amount and $\delta^{18}O$ is evident over central India, the large-scale spatial pattern cannot be explained solely by the classical amount effect.

Instead, the observed isotopic distribution appears to be controlled by the combined influences of large-scale advection of moisture through monsoon circulation and moisture recycling. A phase-space analysis framework is used to examine the interactions between advected atmospheric moisture and evapotranspired moisture and to identify their characteristic timescales. The analysis reveals a strong spatial correspondence between precipitation isotope variability and recycled rainfall, demonstrating that terrestrial evapotranspiration exerts a first-order control on the isotopic composition of precipitation at the continental scale. This finding provides a mechanistic explanation for how moisture recycling shapes precipitation isotope variability across the Indian monsoon region.

The study further shows that isotopically depleted rainfall associated with Bay of Bengal low-pressure systems is partly offset by the continuous advection of relatively enriched moisture from the Arabian Sea, contributing to the observed isotopic patterns over central India. In addition, an updated Indian Meteoric Water Line (IMWL) is derived using an expanded precipitation isotope dataset spanning mainland India and the island territories. Together, these results highlight the importance of land-atmosphere coupling and moisture recycling in controlling precipitation isotope variability and provide a new framework for interpreting isotope-based hydrological and paleoclimatic records from the Indian monsoon region.

Keywords: Precipitation isotopes, isoscape, Indian monsoon, Low-pressure system, Core monsoon zone, Bay of Bengal, Proxy records

1. Introduction

The isotopic composition of precipitation, expressed as stable oxygen ($\delta^{18}O$) and hydrogen ($\delta^{2}H$), varies spatially and temporally due to factors such as temperature, humidity, and precipitation amount. The geo-spatial variability of precipitation isotopes is an essential area of research across various fields of hydrological cycles, such as hydro-meteorology and eco-hydrology. Understanding the geo-spatial variability of precipitation isotopes is important for several reasons. Firstly, it can help us identify the sources of precipitation and track the movement of water through the hydrological cycle (Rozanski et al. 1982; Aggarwal et al. 2016; Vystavna et al. 2020; Bowen et al. 2010). Secondly, it provides insights into the mechanisms that control precipitation formation and distribution, aiding in predicting future changes in precipitation patterns (Gat et al. 2001; Guo et al.2019). Thirdly, it can help us understand the impact of climate change on the hydrological cycle and the environment (Gao et al. 2016; Vystavna et al.2020, 2021). A comprehensive understanding of hydrological processes can help us make informed decisions about water management, agriculture, and other critical issues related to weather and climate change (West et al. 2010).

The spatial distribution of precipitation isotopes, also known as 'isoscapes' or isotope landscapes, has been a topic of interest for many researchers. Notable studies were conducted by Gabriel J. Bowen and his colleagues in North America and other regions. They used isoscapes and compared them with observed data to validate biogeochemical models, estimate

biogeochemical fluxes using isotope mass balance, and assess spatial connectivity across biogeochemical, ecological, and anthropological systems (see Bowen 2010 for a review). Sanchez Murillo et al. (2013) mapped the precipitation isotope variability in Costa Rica. In another study, Sanchez Murillo et al. (2023) demonstrated how to trace precipitation patterns in Mexico using this method. Terzer et al. (2013) presented global isoscapes of hydrogen and oxygen isotopes. Vachon et al. (2007) demonstrated the effect of the changes in the seasonal distribution of precipitation patterns on annually averaged $\delta^{18}O$ in the contiguous USA. Several studies were conducted in China, including the Tibetan Plateau (Kong et al., 2019; Shi et al., 2020).

The isoscape technique has rarely been used in South Asia, especially in India. The Indian region is affected by the boreal summer monsoon, which is crucial for agriculture and the economy. The isoscape technique may offer a different perspective on understanding the monsoon processes. Precipitation isotopic data are often generated in isolation; they lack the temporal and spatial continuity of meteorological data. Moreover, point observations or a few observations limited to a small area reveal the processes that control isotopic variability on a local scale. These processes may or may not apply to a larger/continental scale. For instance, precipitation isotopes usually negatively correlate with rainfall amount (known as the 'amount effect'), but some studies reported a positive correlation in specific areas of India (Yadava et al. 2007). Therefore, a robust network of many sampling stations is needed to develop an understanding of isotopic variations over the Indian subcontinent. Since such a network does not exist in India, a monthly compilation of isotopic data spanning several years, representing well-defined climatology, could be an alternative approach that motivates the India-specific isoscape study.

The synoptic systems are weather disturbances with spatial scales ranging from a few hundred to about 1,000 km and characteristic lifetimes of approximately 2–5 days (Sikka, 1977, 1980). During the Indian summer monsoon, rainfall is governed by several sub-seasonal components, including monsoon low-pressure systems and monsoon intraseasonal oscillations. The evolution and interaction of these components are further modulated by global predictors, such as the El Niño–Southern Oscillation (ENSO), which contribute substantially to the variability and predictability of seasonal monsoon rainfall (e.g., Saha et al., 2019). These weather systems' relative contributions and intensities vary across different areas and affect precipitation isotopes differently. To comprehend their integrated effect on the precipitation isotopic variability, the construction and investigation of the spatial patterns of isotopes are essential.

Only a few studies have discussed the geo-spatial isotopic variability and its applications in India. Hobson et al. (2012) studied the migration of dragonflies from India to Africa via the Maldives Islands. These authors mapped the precipitation δ^2H pattern in India and a few adjoining countries using very limited observed precipitation isotopic data. However, they did not address the climate or environmental controls on the spatial pattern of precipitation isotopic variability. Chidambaram et al. (2009) and Dasgupta et al. (2021) conducted regional isoscape studies to investigate the precipitation variability in Tamil Nadu and the surface runoff in the Himalayan catchments, respectively. In the present study, we have compiled a precipitation isotope database for India to address questions related to monsoon processes within an isoscape framework. We have used monthly isotopic data, focusing on the summer monsoon season, which accounts for most rainfall. We address the following science questions: i) how do

precipitation isotopes vary across India? How does it correspond to the spatial pattern of rainfall variability and the recycled ratio of rainfall? ii) Does the spatial pattern of precipitation isotopes reveal specific synoptic-scale systems? iii) The Arabian Sea and Bay of Bengal produce water vapors with distinct isotopic values. What are their roles in determining the spatial distribution of precipitation isotopes in India? Finally, we present a qualitative analysis comparing the model-derived and the observed precipitation isotopic spatial records.

2. Materials and Methods

2.1 Stable Isotope Data

We have sourced publicly available data for this study. The Global Network of Isotopes in Precipitation (GNIP; IAEA/WMO, 2020) repository, maintained by the International Atomic Energy Agency in Vienna, Austria, provides a comprehensive isotope dataset worldwide. This archive provides substantial monthly data for several Indian cities. Kumar et al. (2010) investigated precipitation isotopic variability across India and provided substantial data from several sites. Apart from these two archives, we have used several other sources. A brief note and a map showing the distribution of the sites are provided in the supplementary information section (Fig. S1). Due to the lack of site rainfall data in many cases, we used the raw isotopic records but not the amount-weighted precipitation isotopic values. Also, we did not make any elevation corrections to the isotopic data. We exclusively used the $\delta^{18}O$ data, as the $\delta^{2}H$ dataset contained significant gaps, precluding the generation of a comparable number of d-excess (defined as: d-excess = $\delta^{2}H - 8 * \delta^{18}O$) values. To avoid introducing significant interpolation errors, we refrained from creating spatial plots for d-excess.

2.2 Spatial plot: *isotope*

We compiled observed precipitation isotope data over India from 2000 to 2024 on a monthly basis to create a spatial distribution of precipitation $\delta^{18}O$, as data prior to 2000 are scarce. We used the Python routine 'Pykrige.ok' to interpolate the data where it was unavailable. We chose a linear mode and ran the variogram, which determines the slope and nugget and defines the dataset's variability within a predefined domain. The model calculates the interpolated values for the specified domain. We then used the numpy meshgrid function to combine the latitude, longitude, and interpolated values into a matrix for plotting contours. This method generates data over both land and ocean. However, we excluded the oceanic region due to limited observational data. In addition, the Isotope-incorporated Global Spectral Model version 2 (IsoGSM2) monthly dataset was used for comparison purposes (Yoshimura et al., 2008).

To mask oceanic data, we used the Basemap toolkit and generated a land–ocean mask with Python's Geopandas library. This process removed ocean points while retaining the landmass outline. A loop then checked each output point, keeping those within the landmass and discarding those outside. The resulting dataset enabled us to construct a spatial $\delta^{18}O$ isoscape.

2.3 Spatial plot: rainfall

For the rainfall analysis, we used the India Meteorological Department (IMD) high spatial resolution (0.25° x 0.25°) long-period daily gridded rainfall dataset for India, spanning 1901–2024, as developed by Pai et al. (2014). Climatologies for rainfall were calculated for 2000–2024, in line with precipitation isotope analysis.

The recycling ratio (RR) is the ratio of precipitation originating from local evapotranspiration to the total precipitation in a region. We used the Weather Research and Forecasting (WRF) model with a water vapor tracer (WVT) module (Insua-Costa et al., 2018) to estimate RR. This model tracks moisture from evapotranspiration and captures its lifecycle until it falls as precipitation. We applied this model across the South Asian region, and evapotranspiration is tagged only for land areas. The initial and boundary conditions (6-hourly) are taken from the ERA5 reanalysis (1979-2021; Hersbach et al., 2023).

2.4 Back trajectory calculations

The HYSPLIT model (Stein et al., 2015) was used to identify the dominant moisture source regions/path and estimate relative moisture contributions from the Arabian Sea (AS) and Bay of Bengal (BoB) to the Core Monsoon Zone (henceforth denoted as CMZ, approximately by an area 18-28 °N, 65-88 °E; vide Rajeevan et al., 2010) of India. Towards this, back trajectories (clusters) were calculated for rainy events in June - July - August (JJA) 2017 - 2019 using 1°×1° NCEP Global Data Assimilation System (GDAS) data (gdas1, accessed via ftp://arlftp.arlhq.noaa.gov/archives /gdas1). Here, we considered the Sagar site (23.82 °N, 78.76 °E, 598 m asl) to represent the CMZ. Back-trajectory clusters were calculated using standard frequency analysis, generating new trajectories every 6 hours for 192 hours at 1000 m. The back-trajectory time was fixed at 8-10 days to account for moisture residence time (van der Ent & Tuinenburg, 2017). The PySplit package (Warner, 2018) was used for the above calculations. To estimate the relative moisture contributions between the Arabian Sea and the Bay of Bengal, the clusters were generated for the JJA period in two ways: i) the first three primary moisture source regions based on the change in total spatial variance, and ii) the first 60 clusters were calculated and plotted, a few of which are from the Bay of Bengal.

2.5 Moisture flux calculations

Characterization of moisture fluxes is widely employed to interpret precipitation isotopic variability. In the Indian context, several investigators have quantified moisture fluxes from site to regional scales (Midhun et al., 2018; Chakraborty et al., 2021; Kathayat et al., 2019; Kumari et al., 2026). To estimate the mean zonal moisture transport from the Arabian Sea and the Bay of Bengal, we computed vertically integrated moisture fluxes (VIMF) over two representative domains: the Arabian Sea region (55°E–75°E, 5°N–20°N) and the Bay of Bengal region (80°E–100°E, 5°N–20°N). These domains broadly capture the westerly and easterly moisture inflow over the Indian landmass. Monthly vertically integrated moisture fluxes (VIMF) were calculated using multi-level specific humidity, zonal and meridional wind components from

the ERA5 reanalysis dataset (Hersbach et al., 2023) using the pressure levels of 1000hpa, 925, 850, 800, 750, 700, 650, 600, 550, 500, 450, 400, 350, 300, 250, 200, 150, and 100 hPa. Climatological monthly means were derived for the period 2000–2024. The sign of the VIMF difference indicates the prevailing flow direction: positive values denote westerly (Arabian Sea–derived) fluxes, while negative values denote easterly (Bay of Bengal–derived) fluxes.

2.6 Phase-space analysis

To investigate the coupled dynamics of rainfall generation, moisture recycling, and precipitation isotopic variability, we employed a phase-space analysis framework. The analysis uses total rainfall (RFtotal) and recycled rainfall fraction (ρ) as the two state variables defining the phase-space. Here, (RFtotal) represents precipitation generated from both advected moisture and surface-evaporated moisture, whereas (ρ) quantifies the fraction of total rainfall originating from terrestrially recycled moisture.

Because precipitation isotopes integrate atmospheric processes over finite moisture residence times, daily-scale variability may not adequately capture the cumulative influence of land–atmosphere feedback. Therefore, both (RFtotal) and (ρ) were aggregated using a 10-day moving average, consistent with the characteristic atmospheric moisture residence time (van der Ent & Tuinenburg, 2017) over the monsoon region. The resulting variables were standardized using z-score normalization to enable comparison of relative variability and to define a common phase-space structure.

The corresponding precipitation $\delta^{18}O$ values were then embedded in the (RFtotal, ρ) phase-space as color-coded markers. This approach allows the isotopic response to be interpreted in terms of evolving hydroclimatic states and provides a dynamical framework to examine transitions between advective and recycling-dominated rainfall regimes.

2.7 Specific cases of local meteoric water lines

As noted earlier, we have refrained from including hydrogen isotope data because it is less available than oxygen isotope data. This constraint restricts our ability to fully re-evaluate the Indian Meteoric Water Line (IMWL) originally proposed by Kumar et al. (2010). Nevertheless, we provide ~12 site-specific LMWLs where both $\delta^{18}O$ and $\delta^{2}H$ data are available. These sites are characterized by daily precipitation isotope records; except for Tezpur and Minicoy, all also include rainfall measurements, enabling the calculation of precipitation-weighted oxygen and hydrogen isotope values. In addition to daily datasets, monthly GNIP records (e.g., New Delhi) were incorporated.

Using this combined dataset, we have developed an interactive map—"Isotopic Snapshots of Monsoon Rainfall (ISMR)"—that displays $\delta^{18}O$ and d-excess variability alongside LMWL for each site. The accompanying HTML interface allows users to click individual sites to view time series

of monthly $\delta^{18}O$ and d-excess, along with the corresponding LMWL. Finally, by integrating all available data, we construct a regional meteoric water line that may serve as an alternative reference to the IMWL. This atlas provides a comprehensive snapshot of precipitation isotope variability across India and its neighboring islands.

3. **Results and Discussion**

3.1 Spatial characterization of precipitation isotopes

Fig. 1 shows the spatial distribution of precipitation $\delta^{18}O$ on a monthly scale from June to September (a) to (d). In June, the southwest Indian region exhibits relatively high $\delta^{18}O$ values, averaging around -3‰. These values progressively decrease approximately along a transect from the southwest to the northeast. The directional pattern of spatial variations in $\delta^{18}O$ also remains consistent across other summer months, with minor regional-scale changes. However, the absolute values show significant differences in the magnitude, as evident in the $\delta^{18}O$ color-coded scale.

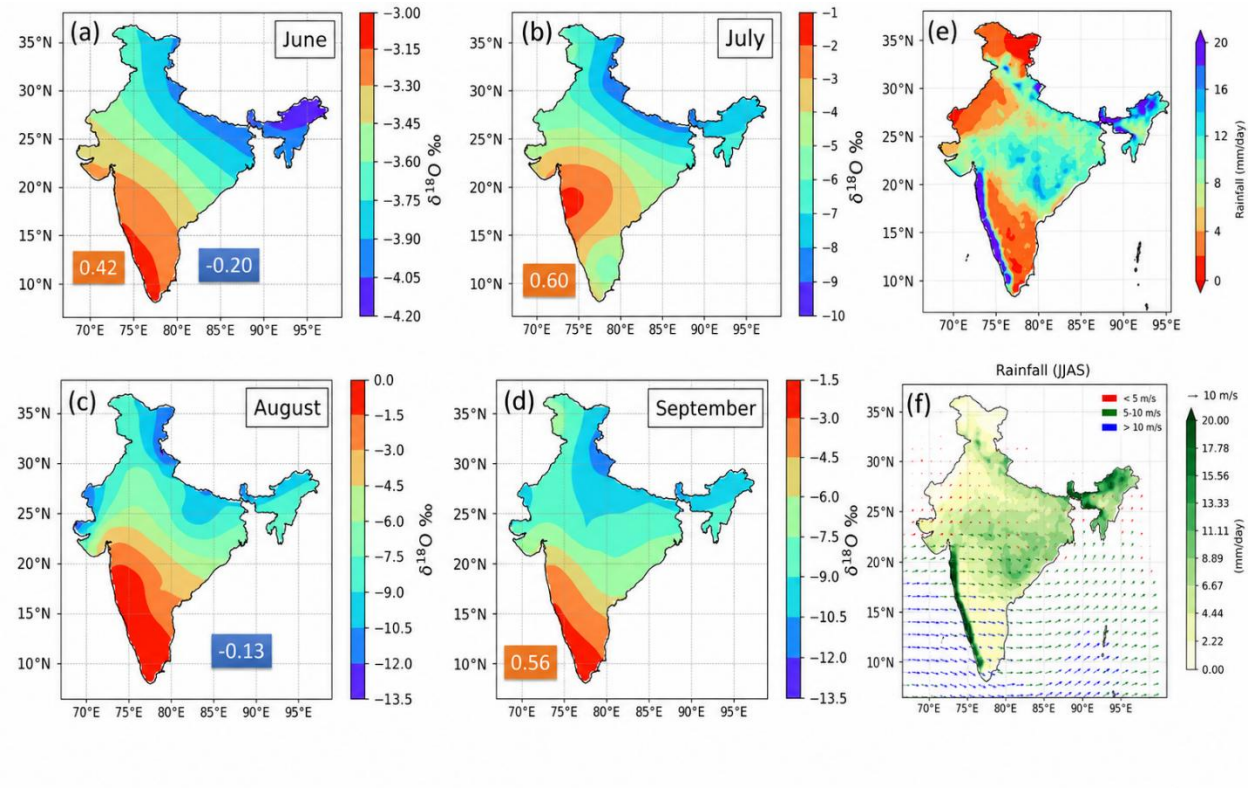


Fig 1. Spatial distribution of $\delta^{18}O$ in India for June (a), July (b), August (c), and September (d). Monthly Climatology (2000-2024) of $\delta^{18}O$ was used. Fig. (b) and [e] compare the spatial distribution of $\delta^{18}O$ and rainfall for July, based on the monthly climatology (2000-2024). An inverse

correlation between rainfall and its isotopic values is evident in the central Indian region. Fig. f shows the spatial distribution of precipitation (shading) and mean wind (vector) for the monsoon (JJAS) season. The surface ocean $\delta^{18}O$ values are shown for the Arabian Sea (orange box) and the Bay of Bengal (blue box) as per the availability of the observed data.

During the monsoon season, the surface waters of the Arabian Sea exhibit relatively high $\delta^{18}O$ values, mostly positive, as indicated by the orange boxes in Fig. 1. Consequently, the vapor and precipitation originating from these waters are isotopically enriched. In contrast, the surface waters of the Bay of Bengal display lower $\delta^{18}O$ values, predominantly below zero, as shown by the blue boxes in Fig. 1. This pattern persists throughout the monsoon season, as evidenced by the climatological $\delta^{18}O$ values presented in blue (boxes) in Fig. 1.

Monsoon circulation is primarily westerly over a major part of the Indian region during boreal summer (Fig. 1f). Moisture carried by this wind undergoes continental effect, progressively losing heavier isotopes and resulting in rainfall with depleted isotopic values. Conversely, the mean $\delta^{18}O$ of the surface waters in the southwest of the Bay of Bengal (BOB) is around -0.3‰ even during the pre-monsoon season (Achyuthan et al., 2010; not shown in the figure). It remains the same within the uncertainty limit of ±0.1‰ during the monsoon season. Therefore, vapor generated from the BOB produces significantly more depleted rainfall than that from the Arabian Sea. Although the BOB-generated moisture contributing to precipitation in India is lower than that from the Arabian Sea (discussed later), the isotopic values of precipitation in India are influenced by both oceanic sources, as reported by several investigators (Sengupta and Sarkar, 2006; Midhun et al., 2018; Ansari et al., 2020).

The isotopic composition of precipitation is strongly influenced by rainfall variability. High rainfall typically results in depleted isotopic values, known as the 'amount effect' (Dansgaard, 1964). While the amount effect is often examined at isolated sites or on a regional scale, Chakraborty et al. (2021) investigated site-specific variability across India. Here, we explore the correspondence between the $\delta^{18}O$-isoscape and spatial rainfall patterns over India.

Fig. 1e and 1b illustrate the spatial distribution of rainfall for July and the corresponding isotopic variability, respectively. The color scheme highlights the relationship between rainfall and isotopic values: blue represents high rainfall and depleted isotopic values, while red indicates low rainfall and higher isotopic values. However, a linear relationship between rainfall and isotopic values is observed only in central India, where green shades in the rainfall map (Fig. 1e) correspond to similar green shades in the isotopic map (Fig. 1b).

The observed correspondence between rainfall and isotopic values in central India suggests that the amount effect is reasonably strong in this region but weak in most other areas. This finding is consistent with previous studies (Breitenbach et al. 2010; Kumar et al. 2010) that have reported weak amount effects in various regions in India.

The amount effect arises from complex interactions among multiple atmospheric processes (Lee and Fung, 2008). One fundamental process is the recycling of moisture, which acts as a secondary moisture source for the cloud system. This secondary moisture can be produced by terrestrial evapotranspiration and raindrop evaporation. Investigating the role of secondary moisture on precipitation isotopes on a spatial scale may provide further insight into the variability of rainwater isotopes. To this end, we have quantified the extent of rainfall derived from secondary moisture sources using the WRF model and examined the spatial distribution patterns of recycled rainfall and precipitation $\delta^{18}O$. Fig. 2 displays the climatological mean monthly recycled rainfall (1979-2020) for June to September. In June, western India exhibits low recycled rainfall values (0-15%), increasing eastward to approximately 25%. By July, the pattern shifts to a southwest-northeast orientation. Notably, this pattern mirrors the spatial variability of $\delta^{18}O$ but in the opposite direction, decreasing from southwest to northeast.

The diagonal pattern of recycled rainfall intensifies in August, with a significant increase in central India. This pattern persists in September, with recycled rainfall peaking in central India. The opposite spatial patterns of recycled rainfall and precipitation $\delta^{18}O$ reveal a mechanistic link between these two parameters.

Interestingly, the spatial pattern of precipitation $\delta^{18}O$ does not show a strong inverse correlation with rainfall amounts (visually or qualitatively), except over central India, consistent with the expected amount effect. In contrast, a clear visual inverse relationship emerges between the spatial pattern of precipitation $\delta^{18}O$ and recycled rainfall. To quantify this, we examined the relationship between area-averaged rainfall (and separately, recycled rainfall) for the CMZ and precipitation-weighted $\delta^{18}O$ on a monthly scale. Supplementary Figure S2 (a and b, respectively) illustrates the linear correlations for these two cases. As evident, area-averaged rainfall exhibits a weak, non-significant correlation ($r = -0.12$, $p=0.54$), whereas area-averaged recycled rainfall shows a moderately strong and highly significant inverse correlation ($r = -0.56$, $p=0.002$) with $\delta^{18}O$. These results clearly indicate that the spatial variability of precipitation isotopes is more strongly governed by recycled rainfall than by primary rainfall. A mechanistic interpretation linking recycled rainfall to isotopic variability is presented in the following section.

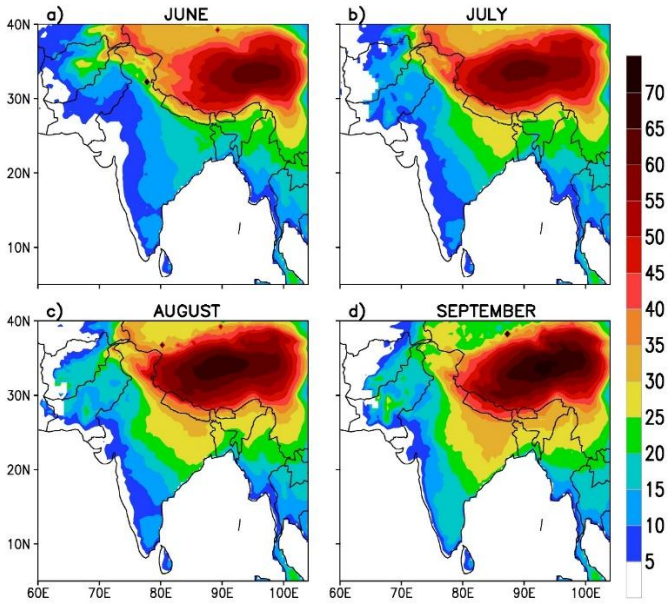


Fig. 2 Monthly climatological mean variability of the recycled rainfall ratio across the land regions for June(a), July(b), August(c), and September(d).

A recent study by Chakraborty et al. (2025) employed statistical analysis, specifically examining the statistical distribution characteristics of precipitation isotopes using histograms. Their findings indicate that the central Indian region, which corresponds to the CMZ, holds promise for proxy-based paleo-monsoon reconstruction.

Notably, the present analysis utilized a different approach – spatial variability of isotopes- which yields a similar conclusion. This convergence of findings from two different approaches lends strong support to our results, underscoring their robustness and reliability.

However, some anomalies exist in the spatial variability maps. For instance, in northeast India, rainfall and isotopic values show a reasonable correspondence. Nevertheless, precipitation isotopic analysis from this region reveals a poor amount effect (Breitenbach et al. 2010; Chakraborty et al. 2022), highlighting the need for further investigation.

3.2 Phase-space analysis

The phase-space trajectory reveals a systematic seasonal evolution of monsoon hydroclimate and its associated isotopic response. During the early monsoon phase, as reflected primarily in Q2 (high ρ, low RFtotal), precipitation exhibits relatively enriched isotopic values. Although the recycled rainfall fraction is high during this phase, the total rainfall amount remains limited, indicating that large-scale monsoon convection has not yet fully intensified. This stage likely reflects residual pre-monsoon moisture and locally evaporated vapor originating from relatively dry land surfaces. Under such conditions, evapotranspiration operates in a relatively isolated manner, producing isotopically enriched precipitation through localized convective recycling.

As the monsoon advances, the trajectory shifts progressively toward Q3 and Q4, reflecting changes in the balance between advected and recycled moisture contributions. Q3 corresponds to a weak rainfall regime where both total rainfall and recycled rainfall fraction remain low. Subsequently, the system enters Q4, characterized by increasing total rainfall, primarily driven by enhanced large-scale moisture advection. Notably, despite increased rainfall during this stage, isotopic values remain moderately enriched or only weakly depleted, indicating that enhanced rainfall alone does not necessarily lead to strong isotopic depletion. This behavior deviates from the classical interpretation of the amount effect, which assumes a direct inverse relationship between rainfall amount and precipitation isotopes.

As monsoon convection intensifies further, the phase-space trajectory evolves toward Q1, where both total rainfall and recycled rainfall fraction are high. This regime exhibits the strongest isotopic depletion, represented by clusters of highly negative $\delta^{18}O$ values. The concentration of points near the Q1–Q2 boundary suggests that maximum isotopic depletion occurs under conditions of sustained recycling combined with moderate-to-high rainfall generated from both advected and terrestrially recycled moisture.

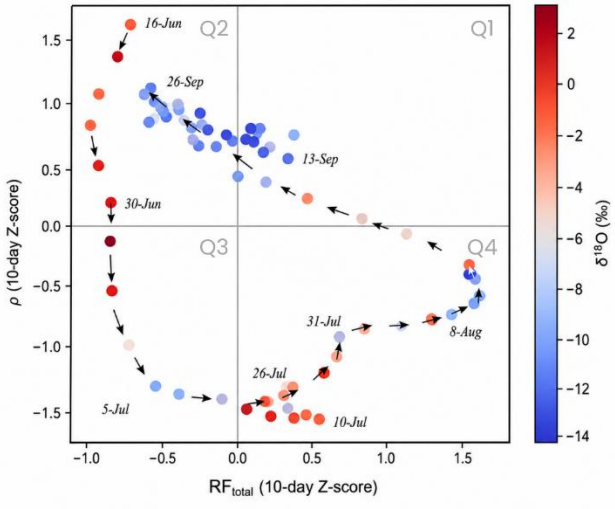


Fig. 3 Phase-space representation of monsoon moisture dynamics during the 2019 monsoon season over the core monsoon zone. The x-axis shows the standardized anomaly (z-score) of total rainfall (RFtotal), and the y-axis shows the standardized recycled rainfall fraction (r), both averaged over a 10-day window. The scattered points are colored by precipitation $\delta^{18}O$ values, with red and blue colors indicating isotopically enriched and depleted precipitation, respectively. Selected dates are labeled to illustrate the seasonal evolution of the monsoon trajectory in phase-space.

The four quadrants represent distinct hydroclimatic regimes: Q1 corresponds to high total rainfall and a high recycled rainfall fraction; Q2 represents high recycled rainfall but relatively low total rainfall; Q3 denotes low total rainfall and low recycled rainfall; and Q4 represents high total rainfall, dominated primarily by advected moisture with a relatively low recycled contribution. Arrows indicate the temporal evolution of the monsoon system through these regimes during the seasonal cycle.

The observed progression demonstrates that isotopic depletion is not controlled solely by rainfall amount, but rather by the cumulative efficiency of moisture recycling and repeated condensation–rainout processes operating over finite atmospheric residence times. During the mature monsoon phase, continuous soil moisture recharge sustains evapotranspiration fluxes, which feed into a progressively recycled atmospheric moisture reservoir. Moisture undergoes repeated condensation and rainout cycles, producing enhanced Rayleigh distillation and increasingly depleted isotopic signatures.

Importantly, the phase-space trajectory forms a quasi-circular closed loop, representing a complete monsoon life cycle in dynamical phase-space. This closed-loop structure indicates that precipitation isotopes respond to the evolving trajectory of the monsoon system rather than to isolated instantaneous processes. The isotopic evolution, therefore, reflects transitions between distinct hydroclimatic regimes governed by varying contributions of advected moisture, recycled rainfall, and moisture residence time.

We also calculated daily mean values of $\delta^{18}O$ from 2016 to 2021 and examined their association with local rainfall and the recycled rainfall ratio over the CMZ. This provides complementary evidence supporting the proposed hypothesis, with results presented in a later section.

3.3 Synoptic processes and precipitation isotopes

The monsoon is characterized by various synoptic-scale processes, including western disturbances and low-pressure systems (LPS), which can significantly influence precipitation isotope variability. Several studies have investigated the impact of these processes on site-specific precipitation isotopic variability (Lone et al. 2020; Jeelani et al. 2018; Chakraborty et al. 2025). However, examining their role in shaping the spatial pattern of isotopic variability may provide additional insights.

Although a dense network of observations is ideal for this purpose, the lack of such a network has necessitated a qualitative interpretation of our findings. Specifically, we focus on the impact of LPS on precipitation isotopes, offering a preliminary understanding of this complex relationship.

The LPSs are the dominant synoptic-scale rain-bearing systems of the South Asian summer monsoon. They typically form over the northern Bay of Bengal and propagate north-westward across the Indian subcontinent, producing a substantial fraction of the seasonal monsoon rainfall (e.g., Saha et al., 2019). Chakraborty et al. (2025) discussed the effect of LPS on precipitation isotopes for a few sites across the CMZ. LPSs usually deplete the isotopic values of rainfall. Since they originate in the northern BOB region, they impact most coastal regions north of the BOB (*op cit.*). As a result, the isotopic values in this region range from -7 to -9‰ (Fig. 1 b, c, d). When the LPSs and, in turn, the moisture travel in a northwestward direction, the isotopic values should progressively decrease due to the continental effect (Cris, 1999). However, as depicted in Fig. 1, the isotopic values remain nearly the same through the corridor extending from the north BOB to west Rajasthan, especially in June and July. This observation may be explained by moisture

dynamics driven by LPS (moisture mainly from the BOB) and by westerly circulations carrying isotopically enriched moisture generated in the Arabian Sea.

3.4 Basin-Scale Moisture Fluxes and Isotope Variability

We have estimated the relative proportions of moisture originating from the BOB and the Arabian Sea that arrive at Sagar, a sample location in the CMZ, for the JJAS periods of 2017, 2018, and 2019. Analysis of the moisture trajectories using the HYSPLIT model shows that under strong southwesterly winds during the boreal summer, significant moisture comes from the Arabian Sea region to CMZ and relatively less from the BOB (Fig. 4, the first numbers in parentheses, in percentage). However, we further partitioned the clusters and found some events (trajectories) when moisture comes from the BOB to CMZ; though the trajectory numbers in these clusters are fewer, the moisture proportion is relatively high, as evident from the specific humidity (q, Fig. 4).

We further estimated the basin-scale moisture fluxes originating from the Arabian Sea and the Bay of Bengal. Fig. 5 presents the monthly climatologies of vertically integrated moisture transport from these two source regions during the southwest monsoon (June–September). The Arabian Sea contribution is consistently larger than that from the Bay of Bengal, peaking in July. Thereafter, the Arabian Sea flux progressively weakens, while the relative contribution from the Bay of Bengal becomes increasingly important. By September, moisture transport from the two basins becomes nearly comparable. This seasonal evolution is noteworthy because the Arabian Sea and Bay of Bengal represent isotopically distinct moisture sources, with Arabian Sea-derived moisture generally being more enriched than moisture originating from the Bay of Bengal.
Viewed in this context, the isotopic composition of precipitation over the CMZ can be interpreted as the net outcome of competing influences from these two moisture reservoirs. LPSs, which primarily originate over the Bay of Bengal, are associated with isotopically depleted precipitation and can substantially lower rainfall $\delta^{18}O$ during individual events. However, their influence is episodic and typically persists for only a few days. In contrast, the south-westerly monsoon flow continuously transports large quantities of relatively enriched Arabian Sea moisture into central India, exerting a persistent background control on precipitation isotopes. The dominance of Arabian Sea moisture during the early monsoon therefore moderates the isotopic depletion expected from repeated LPS activity, helping to explain the relatively weak spatial isotopic gradient along the corridor extending from the northern Bay of Bengal to western Rajasthan (Fig. 1).

As the monsoon progresses, the contribution from the Arabian Sea decreases relative to that from the Bay of Bengal, while recycled continental moisture becomes increasingly important. This seasonal shift provides a plausible explanation for the observed intra-seasonal decline in precipitation $\delta^{18}O$ across central India (Midhun et al. 2018; Chakraborty et al. 2025). To examine whether the seasonal evolution in basin-scale moisture transport alone can explain the observed isotopic depletion, we further investigated the role of continental moisture recycling. Fig. 6 presents the daily climatological evolution (2016–2021) of precipitation $\delta^{18}O$ (red line), local

rainfall (blue bars), and the recycled rainfall ratio (ρ; grey bars, with the axis inverted for clarity) during June–September. During the early monsoon, precipitation $\delta^{18}O$ exhibits only a weak correspondence with the recycled rainfall ratio, consistent with the dominance of Arabian Sea moisture transport (Fig. 5). As the monsoon progresses, however, recycled moisture becomes increasingly important, and $\delta^{18}O$ shows a clear inverse relationship with ρ, a behavior that was also evident from the monthly analysis (Fig. S2b). In contrast, $\delta^{18}O$ displays little systematic relationship with either local rainfall (visually illustrated in Fig. 6) or large-scale rainfall (Fig. S2a), indicating that rainfall amount alone cannot explain the observed seasonal isotopic evolution.

The progressive isotopic depletion from June to September therefore, reflects the combined influence of several processes acting simultaneously. Continued rainout along moisture transport pathways progressively depletes the advected vapor, while the relative contribution of isotopically depleted Bay of Bengal moisture increases as the Arabian Sea contribution weakens (Fig. 5). At the same time, continental moisture recycling becomes progressively stronger, further reinforcing isotopic depletion over the core monsoon zone. Collectively, these processes produce the observed seasonal decline in precipitation $\delta^{18}O$. Because the isotope data analyzed here represent monthly averages, the short-lived isotopic excursions associated with individual low-pressure systems are largely smoothed out, resulting in the comparatively modest month-to-month isotopic variability observed across the central Indian corridor.

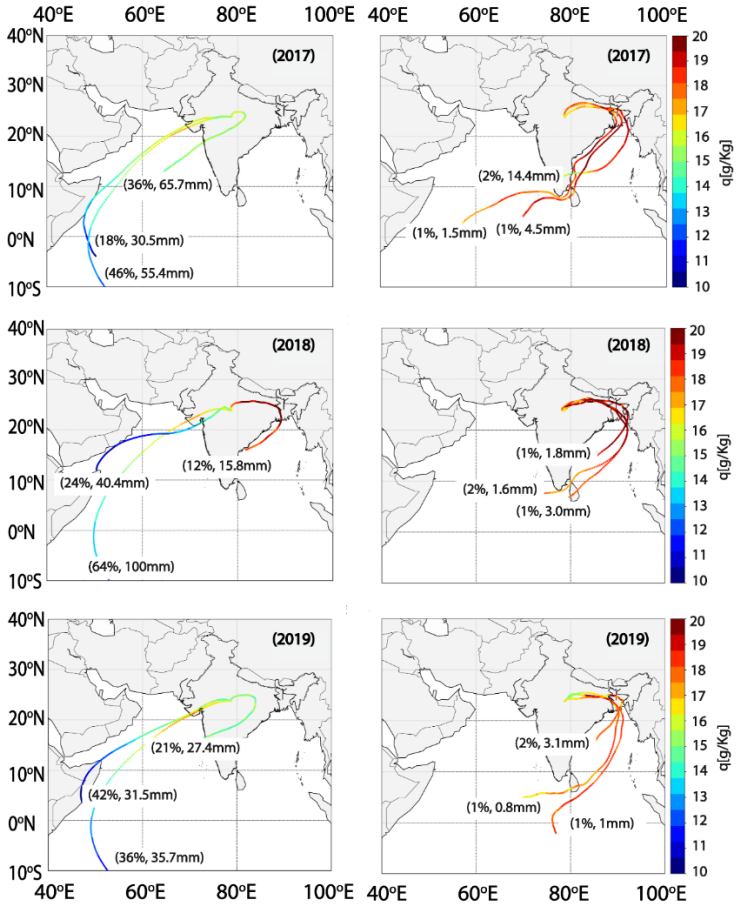


Fig. 4. Hysplit model-derived wind trajectories showing moisture contributions from the Arabian Sea and the Bay of Bengal at an observation site, Sagar, in the core monsoon zone of India. The numbers in the parentheses represent the rainy-day trajectory percentage from the respective seas and the corresponding rainfall amount received at Sagar. The average moisture contribution of the AS is significantly higher than that of the Bay of Bengal.

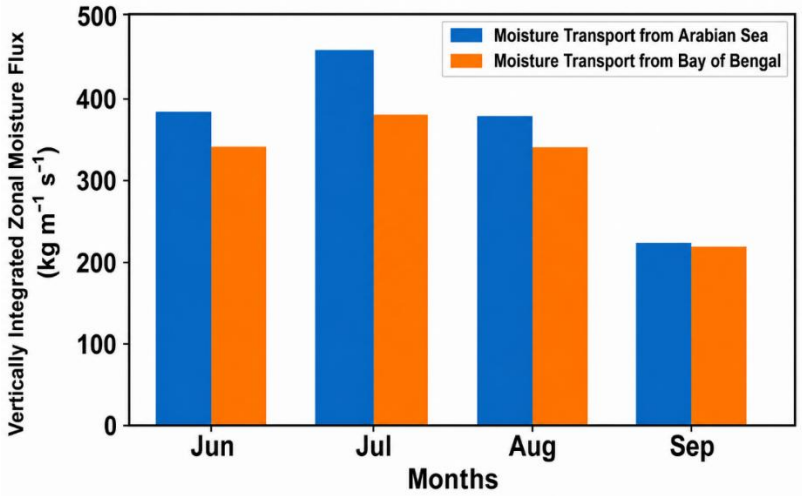


Fig. 5 Monthly climatological vertically integrated moisture transport from the Arabian Sea (blue bars) and the Bay of Bengal (orange bars) associated with precipitation over India during the southwest monsoon (June–September). Basin-scale moisture fluxes are averaged over 2000–2025. The contrasting contributions of these two moisture source regions are used to qualitatively assess their influence on the observed variability of precipitation stable isotopes across India.

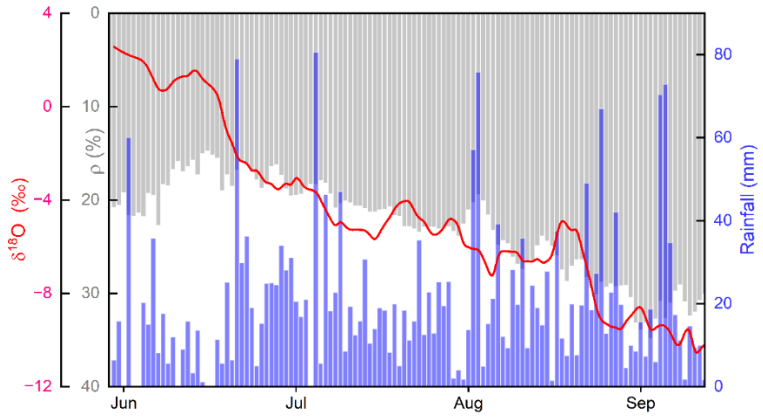


Fig. 6 Time series of the daily mean (2016–2021) of precipitation $\delta^{18}O$ (red line), local rainfall (blue bars), and the recycled rainfall ratio (ρ, grey bars) on an intra-seasonal time scale over the core monsoon zone of India. Note that the recycled rainfall axis is inverted to highlight its inverse correspondence with $\delta^{18}O$.

3.5 The Indian Meteoric Water Line

Meteoric water lines (MWLs) constitute a fundamental framework in isotope hydrology, describing the relationships between δ^2H and $\delta^{18}O$ in precipitation and reflecting the integrated influences of moisture-source conditions, atmospheric transport pathways, and regional hydroclimatic processes. While the Global Meteoric Water Line (GMWL) provides a first-order reference, regional MWLs are essential for capturing spatially distinct climatic and geographical controls on precipitation isotopes. Consequently, they serve as important baselines for groundwater recharge assessments, hydrological tracing, ecohydrological investigations, and paleoclimate reconstructions.

The first Indian Meteoric Water Line (IMWL) was proposed by Kumar et al. (2010) as: $\delta^2H = 7.93\ \delta^{18}O + 9.94$. Using an expanded precipitation isotope dataset spanning approximately 2000–2024, we re-evaluated the IMWL for the Indian region. The revised dataset comprises 430 monthly paired observations of precipitation-weighted δ^2H and $\delta^{18}O$ from twelve monitoring locations distributed across mainland India and the island territories, including Minicoy and the Andaman Islands. The resulting IMWL is:

$$\delta^2H = (7.56 \pm 0.06)\ \delta^{18}O + (7.27 \pm 0.30)$$

Although the revised slope and intercept are slightly lower than those reported by Kumar et al. (2010), we do not interpret these differences as evidence of a systematic shift in the regional hydrological regime. Rather, the updated relationship likely reflects the improved spatial and temporal representativeness afforded by the expanded observational dataset. Therefore, the revised IMWL should be regarded as a more comprehensive characterization of the long-term isotopic behavior of precipitation across India.

To complement the IMWL, we have developed an interactive visualization platform entitled *Isotopic Snapshots of Monsoon Rainfall (ISMR)*. The accompanying HTML application provides site-specific information on precipitation isotope variability across India, including time series of $\delta^{18}O$ and d-excess, together with Local Meteoric Water Lines (LMWLs) derived from monthly observations at each monitoring location. This platform serves as a convenient reference for exploring the spatial and temporal variability of precipitation isotopes across the Indian monsoon domain and may facilitate future hydrological and climatological investigations.

3.6 Comparison of the isoscapes observed versus model simulation of precipitation isotopic records

Our study's spatial distribution of precipitation $\delta^{18}O$ shows good agreement with the pattern reported by Hobson et al. (2012), which was based on a limited number of observed precipitation isotopic data from India and neighboring countries. A visual comparison of these two isoscapes (Fig. S3 in Supplementary Information) reveals a similar pattern. However, the isoscapes prepared using the modeled data show some discrepancies.

We plotted spatial $\delta^{18}O$ patterns for the four monsoon months using IsoGSM2 data (Fig. S4 in Supplementary Information). The results reveal a progressive depletion in oxygen-isotopic values from southwest to northeast, particularly pronounced in the later months (July-September), broadly consistent with observed records. However, regional-scale variability patterns differ from the observed data (Fig. 1 and S4). Previous studies (Nimya et al. 2022) have reported discrepancies between modeled and observed data at specific sites, attributing them to various factors. Explaining the country-scale mismatch, however, requires a comprehensive analysis that is beyond the scope of this study.

3.7 Limitations of the Study/Future Implications

Several factors constrain the spatial mapping analysis of precipitation isotopes presented in this study. Firstly, the data density: a dense network of data points is required for accurate spatial mapping. Unfortunately, such a network is unavailable in India, limiting the study's precision. Secondly, the uneven data distribution- the number of data points varies across monsoon months, with July having the most (94) and other months having significantly fewer (around 60). This uneven distribution may introduce biases in the analysis. The third factor is interpolation errors: interpolation techniques were used to address data gaps. However, these methods can introduce significant errors, especially in regions with sparse data coverage and complex topography. The accuracy of the interpolated data is contingent on the density and distribution of the original data points. Gap filling by interpolation may also be sensitive to the software used.

These limitations highlight the need for a more extensive and dense data network to enable more precise spatial mapping and analysis in future studies. However, despite these limitations, the study has potential.

We have yet to analyze the spatial pattern of d-excess ($\delta^{2}H - 8* \delta^{18}O$), which is a tool to track moisture sources. With improved data availability, a precise pattern of spatial d-excess variability could provide insights into large-scale processes. For instance, it can enhance our understanding of the hydrological cycle, including evaporation and precipitation processes, which is particularly useful in regions with complex water sources and varying climatic conditions (Xiang et al., 2022). The d-excess mapping can also be used for environmental monitoring and assessment, helping to track changes in moisture sources and their impacts on ecosystems. Additionally, understanding moisture sources can aid in agricultural planning and management, especially in monsoon-dependent regions, by providing valuable information for predicting crop yields and planning irrigation. Accurate mapping can support water resource management by identifying precipitation sources and their contributions to local water supplies, crucial for areas prone to water scarcity (Froehlich et al. 2002). Finally, d-excess can serve as an indicator of climate change, reflecting shifts in moisture sources and atmospheric conditions, aiding in the study of global warming and other climatic changes.

4. Conclusions

We constructed a spatially continuous representation of precipitation $\delta^{18}O$ across India to investigate its large-scale variability and to examine the atmospheric and land-surface processes governing its distribution. The major findings of this study are summarized as follows.

1. The spatial distribution of precipitation isotopes does not uniformly follow the classical amount effect across India. Although an inverse relationship between rainfall and $\delta^{18}O$ is evident over the central Indian region, this relationship is absent or considerably weaker in many other parts of the country. Instead, precipitation $\delta^{18}O$ exhibits a pronounced southwest-to-northeast decreasing gradient, with relatively enriched values over southwestern India and progressively depleted values toward northeastern India.

2. The observed isotopic gradient closely mirrors, but in the opposite sense, the spatial distribution of recycled precipitation. Regions with weaker recycling exhibit relatively enriched isotopic values, whereas areas with stronger recycling are associated with more depleted precipitation. Phase-space analysis reveals a mechanistic link between land-surface evaporation, atmospheric moisture recycling, and precipitation isotope variability through the dynamic coupling of advected and evapotranspired moisture operating on different timescales. Importantly, while the role of recycled moisture has long been recognized, this study provides one of the first process-based explanations for the large-scale spatial pattern of precipitation-isotope ratios over India by explicitly linking them to evapotranspiration dynamics.

3. ynoptic-scale weather systems exert an important control on isotopic variability. Low-pressure systems originating over the northern Bay of Bengal typically produce isotopically depleted rainfall. However, their episodic influence is moderated by the persistent transport of relatively enriched moisture from the Arabian Sea. The balance between these two moisture sources helps explain the comparatively weak spatial variability of precipitation isotopes across much of central India and contributes to the observed intra-seasonal evolution of precipitation $\delta^{18}O$ during the monsoon season.

4. An additional outcome of this study is the development of an updated Indian Meteoric Water Line (IMWL), based on an expanded precipitation isotope dataset spanning mainland India and the island territories. The revised IMWL exhibits slightly lower slope and intercept values than the previously reported Indian MWL and the Global Meteoric Water Line. The slope likely reflects the combined influences of sub-cloud evaporation, atmospheric moisture recycling, and the diverse moisture transport pathways associated with the Indian monsoon system. The intercept may indicate the predominance of moisture derived from the humid tropical Indian Ocean, where evaporation generally occurs under conditions of higher ambient humidity than the global mean.

5. The reconstructed spatial pattern of precipitation isotopes agrees well with previous observationally based studies despite the limited spatial coverage of available measurements. However, comparison with isotope-enabled model simulations reveals substantial regional discrepancies, indicating that important physical processes governing

precipitation isotope variability remain inadequately represented. These differences underscore the need for expanded observational networks, long-term isotope monitoring, and improved representation of land-atmosphere interactions and moisture recycling processes in isotope-enabled climate models.

Overall, this study demonstrates that the spatial variability of precipitation isotopes over India cannot be explained solely by rainfall amount. Instead, it emerges from the combined influence of moisture source characteristics, atmospheric transport pathways, synoptic-scale disturbances, land-surface evapotranspiration, and moisture recycling. These findings provide a new process-based framework for interpreting precipitation isotope variability across the Indian monsoon region and offer important implications for hydrological, climatological, and paleoclimatic applications.

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study [word count: 114].

Credit:
DK: Formal analysis, graphics, review, and editing.
FAA: Formal analysis, review, and editing.
CTT: Formal analysis, review, and editing.
NS: Formal analysis, review, and editing.
KK: Visualisation, review, and editing.
SKS: Formal analysis, review, and editing.
PMM: Methodology, review, and editing.
SPS: Methodology, review, and editing
CG: Administrative support, review, and editing.
SC: Conceptualization, formal analysis, manuscript writing, review, and editing.

Data Availability Statement: The data that supports the findings of this study are available in the supporting information of this article. The interactive isotopic map of India is available here.
Also please see the license statement here.

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